



SRIPALEE COLLEGE DIGITAL PLATFORM

Project Proposal Report

A secure, multilingual and accessible government-school website,
historical archive, digital noticeboard, content-management system
and protected academic information portal.

Prepared for project planning and official review

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Executive Summary

This report proposes a modern digital platform for Sripalee College, Horana, designed for the needs and responsibilities of a Sri Lankan government school. The project combines an official public website with secure administration, academic services, historical preservation, multilingual communication and carefully controlled digital innovation.

The proposed platform will improve communication, simplify approved administrative work, support teacher and student record management, strengthen examination workflows and preserve the school's historical identity. Public information and private academic records will be separated so that sensitive student and guardian information is never exposed through the public website.

Recommended delivery approach

Begin with the public website, multilingual content-management system, teacher directory, historical archive and digital noticeboard. Introduce student, parent and examination services only after formal approval, security testing and data-protection procedures are complete.

Key Project Components

Area	Summary
Public website	Official school information, academics, teachers, news, events, clubs, sports, achievements, resources and contact details.
Administration system	Secure CRUD operations, approvals, translations, student and staff management, audit logs and backups.
Academic portal	Protected access for students, parents, teachers and authorised administrators.
Historical archive	Verified timeline, photographs, documents, people, traditions, audio, video and virtual museum content.
Digital services	Progressive Web App, noticeboard, notifications, approved-information AI assistant and secure file storage.

1. Project Objectives

- Present accurate, approved and accessible information about Sripalee College.
- Improve communication with students, parents, teachers, past pupils and the public.
- Manage announcements, events, school documents and staff profiles through a secure CMS.
- Register and manage students using official student IDs and admission numbers.
- Support controlled examination, result-entry, review, approval and publication workflows.
- Preserve the school's cultural and historical identity through a verified digital archive.
- Provide Sinhala, English and Tamil content through a reviewed publication process.
- Maintain strong privacy, security, accountability and long-term maintainability.

2. Government-School Governance

Because Sripalee College is a government school, the platform must be treated as an official institutional system rather than a personally owned website. Formal approval should be obtained before the platform is publicly presented as official or populated with real student and guardian records.

Required ownership and control

- Official domain name and DNS settings
- Hosting and database accounts
- Administrative and school email accounts
- Backup storage and security services

- Source-code repository and deployment credentials
- Analytics and notification accounts

These resources should be owned by the school or the relevant government authority. During development, dummy records should be used until approval is granted for real data, access permissions, hosting, privacy notices and retention procedures.

3. Proposed System Structure

Area	Summary
Public Website	Approved information for students, parents, past pupils and visitors. No private student data.
Admin Dashboard	Content, staff, student, examination, translation, user, audit and backup management.
Private Portal	Timetables, attendance, assignments, results, progress reports and official messages.
Digital Services	PWA, notifications, noticeboard, secure search, file storage and AI knowledge services.

4. Public Website

The public website should be modern, mobile-responsive and uncluttered. It should help visitors find important information quickly while presenting the school's identity with dignity and consistency.

Recommended navigation

- Home
- Our School
- Academics
- School Life
- News and Events
- Heritage
- Resources
- Contact

Homepage priorities

- Official crest and school name
- Sinhala, English and Tamil language selector
- Emergency announcement bar
- Approved main photograph or short video
- Principal's welcome message
- Latest announcements and upcoming events
- Quick links to admissions, notices, calendar, downloads and portal
- Academic, achievement, school-life and heritage previews
- Approved gallery content and official contact information

5. Multilingual and Accessible Design

The interface should support Sinhala, English and Tamil. Important announcements, emergency notices, admissions information, navigation, policies, forms and error messages should be available in all three languages.

Machine translation may assist the workflow, but official content should be reviewed before publication. The design should use Unicode-compatible fonts, clear headings, readable text, strong contrast, consistent spacing and responsive layouts.

Accessibility requirement

The platform should include keyboard-friendly navigation, clear focus states, readable contrast, image descriptions where appropriate and a Reduce Motion option for users who prefer minimal animation.

6. Academic Staff and Teacher Management

Public teacher profiles

- Photograph (optional)
- Full name and position
- Department and subjects taught
- Qualifications
- Short approved introduction
- Clubs or activities supervised

Private staff records and CRUD

- Create, view and update staff records
- Assign departments, subjects, classes and responsibilities
- Upload or replace approved photographs
- Control public visibility
- Mark staff as transferred or retired
- Archive, restore and review record history

Private phone numbers, home addresses, identity information and personal email addresses must not be displayed publicly. Important staff records should normally be archived rather than permanently deleted.

7. Student and Guardian Management

Each student should have an internal system UUID, official student ID and admission number. The internal UUID preserves database integrity, while the school-facing identifiers support daily administration.

Basic student record

- Full name and name with initials
- Student ID and admission number
- Date of birth and admission date
- Academic year, grade, class and medium
- House and current status
- Guardian relationship
- Optional photograph

Secure search

Authorised staff may search by student ID, admission number, name, grade, class, academic year, house or status. Student search must remain inside the protected system. Name searches should show only enough information to distinguish matching students before permission is checked for the full record.

Guardian access

Parents and guardians should receive separate verified accounts linked only to their own children. They may view approved attendance summaries, results, progress reports, assignments and notices, but never records belonging to other students.

8. Academic, Examination and Results Management

The academic structure should manage years, terms, grades, classes, subjects, departments, enrolments and teacher assignments. Results should be linked through the student's academic-year enrolment, class, subject and examination to preserve historical accuracy.

Examination workflow

Workflow

Draft -> Approved -> Open for Mark Entry -> Under Review -> Final Approval -> Published -> Archived

Result-entry controls

- Teachers may enter marks only for assigned subjects, grades and classes.
- The system validates marks and required fields.
- Subject or section heads review entries.
- Academic administrators approve publication.
- Published results are locked.
- Corrections preserve the original mark, corrected mark, reason, requester, approver and timestamp.

9. Historical Archive

The digital historical archive should be one of the platform's most distinctive features. It should preserve verified information and approved digital materials relating to the school's development, people, traditions and achievements.

- Interactive school timeline
- Wilmot A. Perera section
- Rabindranath Tagore section
- Former principals and distinguished teachers
- Historical buildings and school traditions
- Old photographs, magazines and scanned documents
- Newspaper articles and event programmes
- Audio recollections, video interviews and virtual museum content

Every historical item should include its title, description, date or period, source, contributor, verification status, publication permission, copyright status, related people and translated content where available.

10. Digital Noticeboard, PWA and AI Assistant

Digital noticeboard

A separate full-screen interface may operate on televisions or monitors in the main hall, school sections, staff room and library. It may show notices, events, achievements, examination schedules, emergency messages and approved media. Emergency notices should be able to override normal content.

Progressive Web App

The public website should be installable on mobile devices, load quickly, provide approved notifications and offer limited offline access to public content. Private student information should not be stored unnecessarily on personal devices.

Controlled AI school assistant

The assistant should answer only from approved school information such as admissions, history, events, subjects, clubs, sports, downloads and public notices. It must not expose student records, change marks, predict performance, make disciplinary decisions or invent official information.

11. CRUD and Approval Workflows

All important modules should support Create, Read, Update, Archive and Restore. Permanent deletion should be restricted, particularly for staff, student, examination, result and historical records.

Recommended publication workflow

Content lifecycle

Draft -> Submitted for Review -> Changes Requested -> Approved -> Scheduled -> Published -> Archived

This workflow should apply to teacher profiles, news, announcements, events, achievements, galleries, downloads, historical records, noticeboard content, AI knowledge articles and translations.

12. Visual Design and Animation

The public website should use a professional maroon, ivory and white visual identity, with gold as a controlled accent. The final colours should be confirmed by the school.

Area	Summary
Primary colours	Heritage maroon, deep maroon, warm ivory and white.
Accent colour	Ceremonial gold for borders, icons, active states and achievement details.
Animation	Smooth page transitions, section reveals, image reveals and subtle hover depth.
3D use	Homepage hero, historical timeline, virtual museum and selected achievement features.
Avoid 3D in	Results, attendance, data tables, forms, emergency notices and administrative workflows.

Animations should be brief and purposeful. A Reduce Motion option must replace parallax and 3D transitions with simple fades without removing any functionality.

13. User Roles

Area	Summary
School administration	Principal, Deputy Principal, Academic Administrator, Examination Administrator and Section Head.
Teachers	Class Teacher, Subject Teacher, Attendance Officer, Content Editor and Heritage Editor.
Technical roles	System Administrator, Security Administrator, Backup Administrator and Auditor.
Portal users	Student, Parent or Guardian and Teacher.

Access should depend on both the role and the user's assigned responsibility. For example, a subject teacher manages only assigned subjects, a class teacher accesses assigned classes, a student sees only their own record and a parent sees only linked children.

14. Security and Data Protection

- HTTPS across all services
- Multi-factor authentication for privileged users
- Secure password hashing
- Role-based and assignment-based server-side access checks
- Session expiry, rate limiting and account lockout
- Audit logs for important actions
- File validation and malware scanning
- Encrypted backups and restoration testing
- Result locking and approval workflows
- Privacy notices, consent records and incident procedures
- Regular permission reviews and restricted data exports

The backend must return only the information the current user is authorised to view. Sensitive data must never be sent to the browser and hidden only by the visual interface. Real student records must not be stored in personal developer accounts or unauthorised hosting environments.

15. Recommended Technical Architecture

Area	Summary
Frontend	Next.js, React, TypeScript, Tailwind CSS, Framer Motion and selected React Three Fiber features.
Backend	FastAPI, Python, SQLAlchemy, Alembic, PostgreSQL and Redis.
Storage	Separate areas for public media, historical documents, private student files, temporary uploads and backups.
Authentication	School-managed accounts, secure first login, MFA for administrators, session controls and login auditing.
Proposed services	www, portal, admin, noticeboard and API subdomains under the school-controlled domain.

16. Development Roadmap

Phase	Focus	Main Deliverables
1	Approval and governance	Formal approval, school coordinator, ownership, hosting, data fields, privacy and maintenance responsibilities.
2	Research and requirements	Verified history, records review, user roles, workflows, photography plan and content inventory.
3	UX and design	Sitemap, mobile and desktop wireframes, multilingual design, animation prototype and accessibility review.
4	Public website and CMS	Homepage, academics, staff, news, events, clubs, sports, achievements, resources and multilingual publishing.
5	Historical archive	Timeline, people, photographs, documents, audio, video, source tracking and virtual museum.
6	Digital noticeboard	Screen layouts, slide management, schedules, screen groups, emergency override and offline fallback.
7	Student and staff system	Staff records, student registration, guardians, classes, subjects, enrolments, search and permissions.
8	Examinations and results	Exam setup, mark entry, review, approval, publication, reports and amendment history.
9	AI assistant	Approved knowledge base, multilingual answers, privacy restrictions and human escalation.
10	Testing and launch	Functional, mobile, accessibility, performance, security, backup and user-acceptance testing.

17. Recommended Initial Release

- Public school website
- Multilingual content-management system
- Teacher directory
- News, announcements and events
- Clubs, sports and achievements
- Downloads and contact services

- Historical archive foundation
- Digital noticeboard
- Progressive Web App support
- Role-based administration, audit logs and secure backups

The student, parent and examination portal should be introduced after the public platform is stable and all required approvals, permission rules, data-protection procedures and security tests have been completed.

Conclusion

The proposed Sripalee College Digital Platform will be more than an ordinary school website. It will combine official communication, staff management, student administration, examination processing, parent and teacher services, historical preservation, multilingual access and modern digital experiences within a secure government-school structure.

The platform should be developed in controlled phases, prioritising clarity, accessibility, privacy, security and long-term maintenance. Modern animations and selective 3D effects can provide a polished public experience, while administrative and academic workflows remain simple, fast and accountable.

Project definition

Sripalee College Digital Platform - a secure, multilingual and accessible government-school website, historical archive, digital noticeboard, content-management system and protected academic information portal.